

How much do leaky splits, missing external validation and missing calibration cost?

An empirical companion to a PROBAST+AI / TRIPOD+AI appraisal of machine learning for mental-health prediction, on three public actigraphy cohorts

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Code, run manifests and generated tables:

<https://github.com/rogerpanel/cv/tree/main/mlmh-appraisal>

Abstract

Objective. A companion systematic review found that machine-learning (ML) models for mental-health prediction rarely undergo external validation, rarely report calibration, and often use evaluation designs that leak participant identity. We quantified what each practice costs on public data. **Methods.** Three open actigraphy cohorts recorded with the same device: DEPRESJON (23 depressed, 32 healthy controls), PSYKOSE (22 schizophrenia, the same 32 controls) and HYPERAKTIV (45 ADHD, 40 clinical controls). One calendar day of minute-level activity was the analysis unit (1,818 unique valid days from 162 participants); 31 statistical and circadian features fed a majority-class baseline, logistic regression, random forest and XGBoost with hyper-parameters fixed a priori. E1 compared record-wise with subject-wise five-fold cross-validation on identical data and seeds. E2 trained on one cohort and evaluated the frozen model on another, with the shared controls assigned to exactly one cohort. E3 reported Brier score, calibration slope and intercept, expected calibration error (ECE) and reliability curves for every model, plus a logistic recalibration arm. Ten seeds; subject-level BCa bootstrap intervals; every run traceable to a manifest. **Results.** Record-wise splitting inflated window-level AUROC by 0.065 (95% CI 0.024 to 0.122) to 0.114 (0.060 to 0.181) in DEPRESJON, 0.020 (0.006 to 0.041) to 0.044 (0.017 to 0.080) in PSYKOSE, and 0.130 (0.095 to 0.167) to 0.169 (0.135 to 0.206) in HYPERAKTIV, where the honest estimate was at chance (0.45 to 0.46) and the leaky one reached 0.60 to 0.65 at subject level. Flexible models inflated more than logistic regression in every cohort. Transfer from DEPRESJON to PSYKOSE preserved discrimination (AUROC 0.77 to 0.78) but transfer in the opposite direction reduced AUROC by 0.12 to 0.20, with calibration slopes of 0.27 to 0.63 and intercepts of 0.7 to 1.4; a label-shift transfer to HYPERAKTIV fell to chance. Under honest internal validation, slopes were 0.53 to 0.95 and ECE 0.06 to 0.22; recalibration on the training cohort did not repair the external miscalibration. Subject-level events per candidate predictor were 0.6 to 0.8. **Conclusions.** The three practices the review found to be common each change conclusions on real data, and in the direction of over-optimism. Reporting subject-wise performance, calibration and subject-level sample sizes should be a condition of publication.

1 Introduction

Machine-learning models that classify mental-health status from wearable, speech, text or clinical data are published faster than they are appraised. The companion review [1] applied PROBAST+AI [2] and TRIPOD+AI [3] across the field and counted how often three practices

occur: evaluation designs in which repeated measures from one participant appear on both sides of a train/test split, absence of any external validation, and absence of any calibration assessment. Counting is not costing. A reader of the review can still ask whether a record-wise split matters in practice, whether a model that discriminates well internally will discriminate at all elsewhere, and whether miscalibration is a real problem for models that are only ever used to rank.

The published literature on the cohorts used here illustrates the problem. On DEPRESJON the dataset authors reported a leave-one-patient-out F1 of 0.73 [4]; subsequent papers report accuracies of 0.85 to 0.96 using day-level or “modified” splits [8, 9]; on PSYKOSE, day-level deep models report accuracies near 0.94 [10]. None reports calibration; none validates across cohorts; and, because the PSYKOSE control group is the DEPRESJON control group, any paper that uses both cohorts without noticing has scored half of its test set on training participants.

We designed three experiments, each answering one review outcome, on cohorts chosen so that the measurement device is constant and only the population changes: (E1) the optimism from record-wise splitting, measured as a paired difference on identical data; (E2) the loss from skipping external validation, measured by freezing a model and applying it to another cohort; (E3) the calibration that would have gone unreported, for every model in E1 and E2. Secondary analyses report sex-stratified performance (E4) and a transdiagnostic and five-class arm on the harmonised OBF-Psychiatric release (E5). Our own methods follow TRIPOD+AI item by item (Supplementary Table S1), because a study that scores others cannot be exempt.

2 Methods

2.1 Source of data and participants

DEPRESJON [4], PSYKOSE [5] and HYPERAKTIV [6] are open datasets from Haukeland University Hospital and collaborating Norwegian sites, all recorded with an Actiwatch (Cambridge Neurotechnology) at one-minute resolution, and re-released in harmonised form as OBF-Psychiatric [7]. DEPRESJON contains 23 in- and outpatients with unipolar or bipolar depression (MADRS at start of recording, median 24) and 32 healthy controls; PSYKOSE 22 inpatients with schizophrenia (BPRS median 51) and the same 32 controls; HYPERAKTIV 51 adults with ADHD and 52 clinical controls (psychiatric outpatients without ADHD), of whom 45 and 40 have activity recordings. Recruitment, consent and clinical assessment are described in the source publications. Table 1 summarises the analysed participants. Data were obtained from the dataset authors’ download pages and Zenodo; SHA-256 checksums of every file are recorded in the repository.

2.2 Outcome

The outcome was diagnostic group membership: depression versus healthy control (DEPRESJON), schizophrenia versus healthy control (PSYKOSE), ADHD versus clinical control (HYPERAKTIV). Labels are participant-level; every window inherits its participant’s label, and the data schema rejects a participant with more than one label.

2.3 Predictors

The analysis unit was one calendar day (00:00 to 23:59) of activity counts with at least 80% valid minutes; partial first and last days were dropped. Thirty-one predictors were computed per day (Supplementary Table S2): distributional statistics (mean, SD, coefficient of variation, median, 10th and 90th percentiles, maximum, skewness, kurtosis, proportion of zero-count minutes, mean and SD of $\log(1 + x)$), day/night contrasts (08:00 to 22:00 versus night), non-parametric circadian measures on the hourly profile (M10, L5, relative amplitude, intradaily variability, sine

and cosine of M10 onset), autocorrelation at 5 and 60 minutes, spectral summaries, the number of rest/activity transitions and the missing-minute fraction. No demographic variable was used, so that E2 compares populations rather than covariate availability. A raw representation (the $\log(1 + x)$ 1440-minute series) is available for a 1D convolutional network; its results are reported as a supplementary arm.

2.4 Sample size

No sample-size calculation was possible; the cohorts are fixed. We report subject-level events per candidate predictor (minority-class participants divided by 31) for every training cohort, which is 0.6 to 0.8, and we report window-level counts alongside, because that is the number most papers present.

2.5 Missing data

Days below the validity threshold were excluded (no participant lost all days). Remaining missing minutes were ignored in feature computation; missing feature values were imputed with the training-fold median inside the pipeline.

2.6 Models and pipeline

Majority-class baseline, L_2 -regularised logistic regression ($C = 1$, class-balanced), random forest (500 trees, minimum leaf 2, balanced subsampling) and XGBoost (300 rounds, depth 3, learning rate 0.05, subsampling 0.8). Hyper-parameters were fixed a priori at library defaults or conservative values and were not tuned, so that E1 isolates the splitting design. Imputation and standardisation were fitted on the training fold only inside an `imblearn` pipeline; no resampling was applied in the reported runs.

2.7 Internal validation: E1

Five-fold cross-validation repeated over ten seeds under two splitters on identical data: *subject-wise* (stratified group k -fold; no participant on both sides of any fold) and *record-wise* (stratified k -fold over days, ignoring participant identity). An automated test fails the build if any non-E1 configuration requests the record-wise splitter, and the runner asserts fold disjointness at run time. The inflation is the paired difference (record-wise minus subject-wise) on the same participants.

2.8 External validation: E2

For each ordered pair of cohorts the model was fitted on all windows of cohort A and applied, frozen, to cohort B. The internal comparator is subject-wise cross-validation on A. Because DEPRESJON and PSYKOSE share their 32 controls (verified by hashing every participant’s activity series, not by documentation), each shared control was assigned at random (seed 0) to exactly one cohort, leaving 16 controls on each side; the assignment table is in the repository. The pair DEPRESJON $\rightarrow$ HYPERAKTIV is a deliberate label-shift test (depression model applied to ADHD versus clinical controls) and is expected to fail.

2.9 Performance measures: E3

Discrimination: AUROC (primary), AUPRC, accuracy, balanced accuracy, macro-F1 and MCC at a 0.5 threshold. Calibration: Brier score, calibration slope (logistic regression of outcome on

the logit of the predicted probability) and intercept (with slope fixed at 1) [11], ECE with ten equal-width bins, and reliability curves. All measures were computed at *window* level (the unit most papers report) and at *subject* level (mean predicted probability per participant). Point estimates are means over seeds; 95% intervals are bias-corrected and accelerated bootstrap intervals from 1,000 resamples of *participants* (all of a participant’s windows drawn together), with a leave-one-participant-out jackknife for the acceleration term.

2.10 Recalibration arm

For each E2 pair, a logistic recalibration (slope and intercept) was fitted on the training cohort’s out-of-fold predictions and applied to the external predictions, to test whether the external miscalibration is repairable without access to the target cohort.

2.11 Fairness: E4

Subject-wise E1 predictions were stratified by sex (from the harmonised OBF-Psychiatric metadata) and AUROC, Brier, slope and ECE were recomputed per stratum with subject-level intervals.

2.12 Open science

Every run writes a manifest with git commit, configuration hash, seeds, package versions and dataset checksums; tables and figures in this paper are generated by the code from those runs. The repository is public under an MIT licence; the data are available from their authors under the terms cited.

3 Results

3.1 Participants and windows

Table 1: Participant characteristics per cohort. HYPERAKTIV controls are psychiatric patients without ADHD (clinical controls); DEPRESJON and PSYKOSE share the same 32 healthy controls. Sex, age band and recording days from the harmonised OBF-Psychiatric metadata; day-windows are the analysis units after the 80% validity threshold.

Cohort	Group	n	Female, n (%)	Age (modal band)	Recording days, median (range)	Valid day-windows	Clinical scale, median (IQR)
depresjon	cases (depression)	23	10 (43)	45-49	13 (5-18)	405	MADRS 24 (IQR 18-26)
depresjon	controls (healthy_control)	32	20 (62)	50-54	13 (8-20)	739	–
psykose	cases (schizophrenia)	22	3 (14)	35-39	13 (12-14)	390	BPRS 51 (IQR 44-58)
psykose	controls (healthy_control)	32	20 (62)	50-54	13 (8-20)	739	–
hyperaktiv	cases (adhd)	45	21 (47)	30-39	7 (2-10)	343	ASRS 49 (IQR 40-55)
hyperaktiv	controls (clinical_control)	40	20 (50)	17-29	7 (6-10)	329	ASRS 34 (IQR 25-43)
obf_psychiatric	cases (adhd, clinical, depression, schizophrenia)	130	54 (42)	30-39	7 (2-18)	1467	–
obf_psychiatric	controls (control)	32	20 (62)	50-54	13 (8-20)	739	–

DEPRESJON yielded 1,034 valid days (median 15 per participant), PSYKOSE 1,021 (16), HYPERAKTIV 502 (6); no participant was excluded. Series hashing confirmed that the 32 PSYKOSE control recordings are byte-identical to the 32 DEPRESJON control recordings and that OBF-Psychiatric contains all 162 participants exactly once.

3.2 E1: the cost of leaky splitting

Record-wise splitting raised window-level AUROC in every cohort and for every informative model (Table 2, Figure 1). In DEPRESJON, XGBoost rose from 0.745 (95% CI 0.675 to 0.842) to 0.872 (0.830 to 0.918), a paired inflation of 0.114 (0.060 to 0.181); at subject level the leaky estimate

Table 2: E1: discrimination under subject-wise versus record-wise cross-validation. Cells give the mean over seeds with subject-level BCa 95% bootstrap intervals; inflation is the paired record-wise minus subject-wise difference.

Cohort	Model	Window AUROC subject-wise	Window AUROC record-wise	Window inflation	Subject AUROC subject-wise	Subject AUROC record-wise	Subject inflation
depresjon	Majority class	0.482 [0.292, 0.653]	0.499 [0.459, 0.528]	0.026 [-0.147, 0.218]	0.433 [0.300, 0.551]	0.483 [0.295, 0.555]	0.001 [-0.191, 0.202]
depresjon	Logistic regression	0.765 [0.675, 0.861]	0.840 [0.781, 0.896]	0.065 [0.024, 0.122]	0.888 [0.822, 0.969]	0.961 [0.927, 0.990]	0.065 [0.011, 0.109]
depresjon	Random forest	0.762 [0.681, 0.848]	0.874 [0.823, 0.923]	0.107 [0.058, 0.168]	0.865 [0.812, 0.948]	0.977 [0.949, 1.000]	0.094 [0.041, 0.146]
depresjon	XGBoost	0.745 [0.675, 0.842]	0.872 [0.830, 0.918]	0.114 [0.060, 0.181]	0.862 [0.795, 0.947]	0.983 [0.964, 1.000]	0.111 [0.040, 0.175]
psykose	Majority class	0.476 [0.267, 0.632]	0.499 [0.470, 0.535]	0.062 [-0.113, 0.241]	0.466 [0.278, 0.557]	0.501 [0.331, 0.593]	0.043 [-0.148, 0.227]
psykose	Logistic regression	0.906 [0.820, 0.942]	0.926 [0.875, 0.955]	0.020 [0.006, 0.041]	0.975 [0.943, 1.000]	0.994 [0.983, 1.000]	0.020 [0.000, 0.043]
psykose	Random forest	0.889 [0.798, 0.934]	0.936 [0.890, 0.962]	0.044 [0.017, 0.080]	0.953 [0.913, 0.997]	0.993 [0.983, 1.000]	0.038 [0.000, 0.074]
psykose	XGBoost	0.899 [0.810, 0.941]	0.942 [0.907, 0.969]	0.042 [0.016, 0.078]	0.967 [0.935, 1.000]	0.997 [0.991, 1.000]	0.028 [0.000, 0.059]
hyperaktiv	Majority class	0.496 [0.355, 0.628]	0.496 [0.433, 0.533]	-0.010 [-0.170, 0.148]	0.500 [0.398, 0.597]	0.482 [0.349, 0.540]	-0.054 [-0.203, 0.106]
hyperaktiv	Logistic regression	0.463 [0.360, 0.535]	0.577 [0.486, 0.661]	0.130 [0.095, 0.167]	0.445 [0.328, 0.519]	0.601 [0.502, 0.695]	0.177 [0.131, 0.221]
hyperaktiv	Random forest	0.447 [0.361, 0.534]	0.603 [0.526, 0.662]	0.169 [0.135, 0.206]	0.425 [0.333, 0.518]	0.640 [0.550, 0.722]	0.217 [0.177, 0.257]
hyperaktiv	XGBoost	0.449 [0.360, 0.524]	0.586 [0.518, 0.664]	0.151 [0.116, 0.186]	0.438 [0.337, 0.530]	0.645 [0.555, 0.732]	0.214 [0.173, 0.261]

Table 3: E1: window-level discrimination and calibration for every model, cohort and splitting design (mean over seeds).

Cohort	Model	Split	Accuracy	Macro-F1	AUROC	Brier	Cal. slope	Cal. intercept	ECE
depresjon	Majority class	subject-wise	0.653	0.395	0.482	0.227	-4.01	-0.00	0.000
depresjon	Majority class	record-wise	0.653	0.395	0.499	0.227	0.29	0.00	0.000
depresjon	Logistic regression	subject-wise	0.722	0.706	0.765	0.194	0.52	-0.54	0.118
depresjon	Logistic regression	record-wise	0.761	0.750	0.840	0.163	0.90	-0.62	0.095
depresjon	Random forest	subject-wise	0.724	0.679	0.762	0.185	0.69	0.01	0.061
depresjon	Random forest	record-wise	0.806	0.782	0.874	0.136	1.23	-0.11	0.039
depresjon	XGBoost	subject-wise	0.722	0.676	0.745	0.202	0.43	0.29	0.123
depresjon	XGBoost	record-wise	0.802	0.776	0.872	0.137	0.82	0.07	0.039
psykose	Majority class	subject-wise	0.661	0.398	0.476	0.224	-3.95	0.00	0.000
psykose	Majority class	record-wise	0.661	0.398	0.499	0.224	0.31	0.00	0.000
psykose	Logistic regression	subject-wise	0.856	0.842	0.906	0.115	0.90	-0.60	0.086
psykose	Logistic regression	record-wise	0.868	0.855	0.926	0.103	1.07	-0.62	0.078
psykose	Random forest	subject-wise	0.850	0.829	0.889	0.118	0.95	-0.12	0.048
psykose	Random forest	record-wise	0.878	0.861	0.936	0.093	1.32	-0.17	0.048
psykose	XGBoost	subject-wise	0.862	0.842	0.899	0.108	0.71	0.15	0.046
psykose	XGBoost	record-wise	0.891	0.876	0.942	0.082	0.91	0.07	0.026
hyperaktiv	Majority class	subject-wise	0.504	0.335	0.496	0.250	0.00	0.00	0.000
hyperaktiv	Majority class	record-wise	0.504	0.335	0.496	0.250	0.00	0.00	0.000
hyperaktiv	Logistic regression	subject-wise	0.461	0.460	0.463	0.286	-0.21	-0.01	0.160
hyperaktiv	Logistic regression	record-wise	0.543	0.543	0.577	0.252	0.21	0.01	0.074
hyperaktiv	Random forest	subject-wise	0.463	0.462	0.447	0.286	-0.32	0.00	0.156
hyperaktiv	Random forest	record-wise	0.571	0.571	0.603	0.245	0.57	0.00	0.055
hyperaktiv	XGBoost	subject-wise	0.460	0.460	0.449	0.330	-0.15	0.00	0.248
hyperaktiv	XGBoost	record-wise	0.557	0.557	0.586	0.271	0.26	0.00	0.151

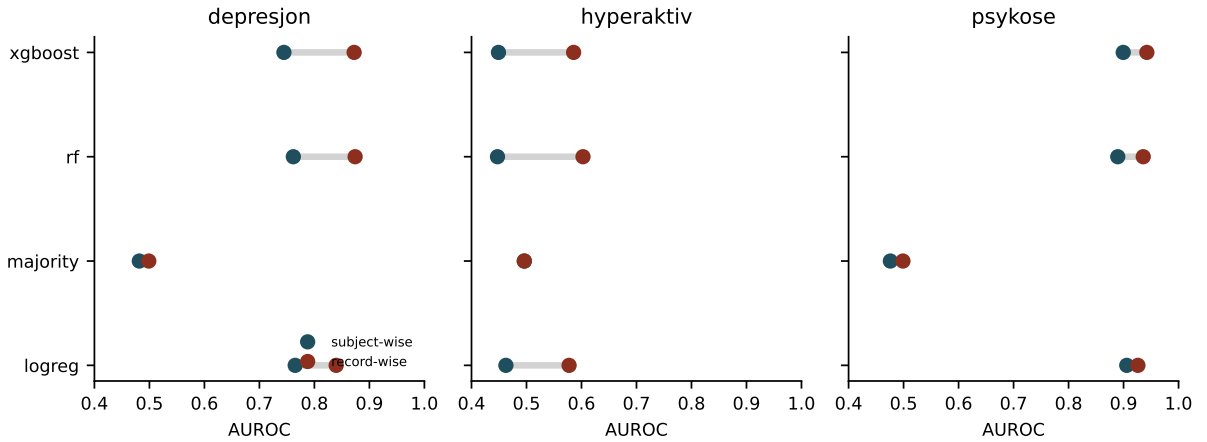


Figure 1: E1. Window-level AUROC of each model under subject-wise (dark) and record-wise (red) five-fold cross-validation, mean over ten seeds. The grey bar is the leakage inflation.

reached 0.983. In PSYKOSE, where the honest estimate is already high (0.899 to 0.906), the inflation was 0.020 to 0.044. In HYPERAKTIV the honest estimate was at or below chance for all three models (0.447 to 0.463 window level; 0.425 to 0.445 subject level), yet record-wise splitting produced 0.577 to 0.603 at window level and 0.601 to 0.645 at subject level, an inflation of 0.130 to 0.169 with intervals excluding zero. Accuracy and macro-F1 behaved the same way (Table 3). Logistic regression inflated least in every cohort (0.065, 0.020, 0.130) and the tree ensembles most.

3.3 E2: the cost of skipping external validation

Table 4: E2: internal (subject-wise CV on the training cohort) versus external (unchanged model applied to the second cohort) performance. Window-level estimates, mean over seeds, subject-level BCa 95% intervals. EPV = minority-class subjects per candidate predictor in the training cohort.

Train → Test	Model	EPV	Internal AUROC	External AUROC	Δ	Int. slope	Ext. slope	Ext. intercept	Ext. Brier
depresjon → psykose	Majority class	0.6	0.475 [0.224, 0.662]	0.500 [0.500, 0.500]	+0.025	-4.16	0.00	-0.02	0.250
depresjon → psykose	Logistic regression	0.6	0.790 [0.701, 0.871]	0.779 [0.676, 0.847]	-0.011	0.66	0.50	0.06	0.193
depresjon → psykose	Random forest	0.6	0.730 [0.607, 0.837]	0.775 [0.688, 0.846]	+0.045	0.58	0.80	-0.02	0.193
depresjon → psykose	XGBoost	0.6	0.716 [0.608, 0.825]	0.768 [0.678, 0.835]	+0.051	0.34	0.55	-0.01	0.204
psykose → depresjon	Majority class	0.6	0.463 [0.206, 0.601]	0.500 [0.500, 0.500]	+0.037	-4.07	0.00	0.06	0.250
psykose → depresjon	Logistic regression	0.6	0.881 [0.790, 0.938]	0.689 [0.564, 0.793]	-0.192	0.72	0.27	1.20	0.262
psykose → depresjon	Random forest	0.6	0.851 [0.756, 0.926]	0.728 [0.620, 0.818]	-0.123	0.76	0.63	0.74	0.225
psykose → depresjon	XGBoost	0.6	0.871 [0.793, 0.936]	0.670 [0.569, 0.770]	-0.201	0.55	0.30	1.44	0.285
depresjon → hyperaktiv	Majority class	0.8	0.482 [0.292, 0.653]	0.500 [0.500, 0.500]	+0.018	-4.01	-0.01	0.65	0.275
depresjon → hyperaktiv	Logistic regression	0.8	0.765 [0.675, 0.861]	0.442 [0.348, 0.532]	-0.323	0.52	-0.16	-0.42	0.350
depresjon → hyperaktiv	Random forest	0.8	0.762 [0.681, 0.848]	0.460 [0.360, 0.557]	-0.301	0.69	-0.15	0.33	0.317
depresjon → hyperaktiv	XGBoost	0.8	0.745 [0.675, 0.842]	0.459 [0.358, 0.550]	-0.286	0.43	-0.09	0.48	0.364

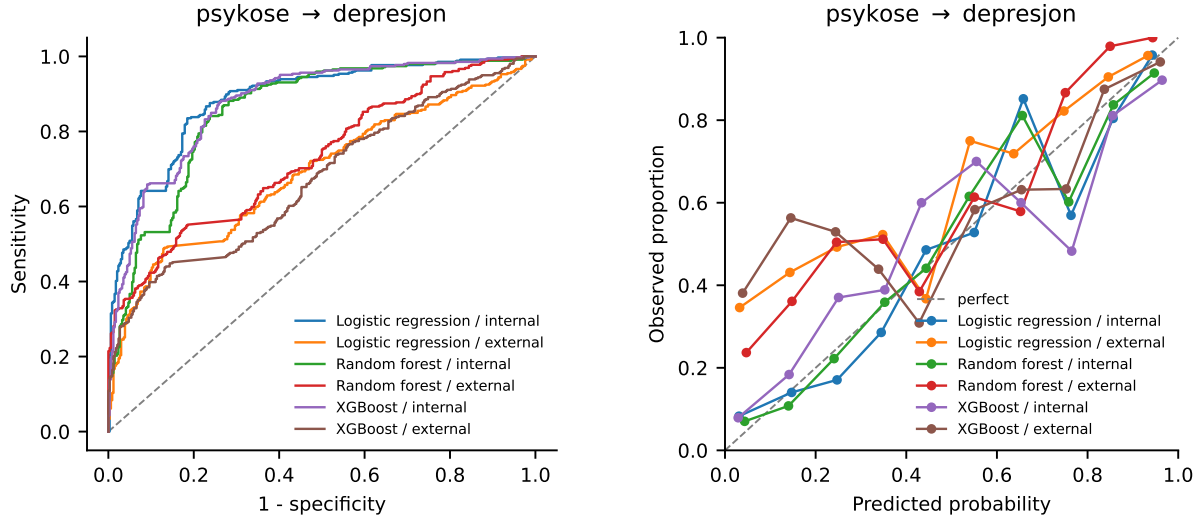


Figure 2: E2, PSYKOSE → DEPRESJON. Left: ROC curves of the internal (subject-wise CV on PSYKOSE) and external (frozen model on DEPRESJON) predictions. Right: reliability curves of the same predictions.

Transfer was asymmetric (Table 4). Models trained on DEPRESJON discriminated schizophrenia from the held-out controls as well as they had discriminated depression internally (external AUROC 0.768 to 0.779 versus internal 0.716 to 0.790), because schizophrenia produces the larger activity deficit. Models trained on PSYKOSE lost 0.12 to 0.20 AUROC when applied to DEPRESJON (logistic regression 0.881 → 0.689; XGBoost 0.871 → 0.670), and their probabilities were both overconfident and shifted: calibration slopes 0.27 to 0.63, intercepts 0.74 to 1.44, Brier 0.22 to 0.29 against 0.25 for the constant prediction (Figure 2). The label-shift pair DEPRESJON → HYPERAKTIV fell to 0.442 to 0.460 with negative slopes. Subject-level EPV in the training cohorts was 0.6 (DEPRESJON, PSYKOSE after the control split) and 0.8 (HYPERAKTIV).

158 3.4 E3: the calibration blind spot

Table 5: E3: calibration alongside discrimination for every model in E1 (subject-wise arm) and E2, plus logistic recalibration of external predictions.

Exp.	Run	AUROC	Brier	Cal. slope	Cal. intercept	ECE
E1	depresjon.logreg.subject_wise	0.779	0.187	0.69	-0.51	0.102
E1	depresjon.rf.subject_wise	0.770	0.181	0.77	0.01	0.056
E1	depresjon.xgboost.subject_wise	0.764	0.192	0.53	0.27	0.105
E1	hyperaktiv.logreg.subject_wise	0.449	0.279	-0.28	-0.01	0.161
E1	hyperaktiv.rf.subject_wise	0.443	0.281	-0.45	0.00	0.137
E1	hyperaktiv.xgboost.subject_wise	0.442	0.313	-0.23	0.00	0.221
E1	psykose.logreg.subject_wise	0.907	0.112	0.95	-0.59	0.088
E1	psykose.rf.subject_wise	0.893	0.115	1.03	-0.12	0.052
E1	psykose.xgboost.subject_wise	0.903	0.103	0.78	0.14	0.052
E2	depresjon-to-hyperaktiv.logreg.external	0.442	0.350	-0.16	-0.42	0.291
E2	depresjon-to-hyperaktiv.rf.external	0.461	0.317	-0.15	0.33	0.214
E2	depresjon-to-hyperaktiv.xgboost.external	0.459	0.363	-0.09	0.47	0.292
E2	depresjon-to-psykose.logreg.external	0.779	0.193	0.50	0.06	0.079
E2	depresjon-to-psykose.rf.external	0.776	0.193	0.81	-0.02	0.044
E2	depresjon-to-psykose.xgboost.external	0.769	0.202	0.56	-0.01	0.085
E2	depresjon.logreg.internal	0.779	0.187	0.69	-0.51	0.102
E2	depresjon.rf.internal	0.770	0.181	0.77	0.01	0.056
E2	depresjon.xgboost.internal	0.764	0.192	0.53	0.27	0.105
E2	psykose-to-depresjon.logreg.external	0.689	0.262	0.27	1.20	0.202
E2	psykose-to-depresjon.rf.external	0.728	0.225	0.63	0.74	0.149
E2	psykose-to-depresjon.xgboost.external	0.671	0.284	0.31	1.43	0.229
E2	psykose.logreg.internal	0.885	0.137	0.77	-0.10	0.061
E2	psykose.rf.internal	0.854	0.151	0.82	-0.08	0.055
E2	psykose.xgboost.internal	0.877	0.148	0.62	-0.16	0.083
E2	depresjon-to-hyperaktiv.logreg.external+recal	0.442	0.311	-0.23	0.22	0.223
E2	depresjon-to-hyperaktiv.rf.external+recal	0.461	0.301	-0.20	0.37	0.176
E2	depresjon-to-hyperaktiv.xgboost.external+recal	0.459	0.311	-0.17	0.36	0.194
E2	depresjon-to-psykose.logreg.external+recal	0.779	0.211	0.72	0.56	0.139
E2	depresjon-to-psykose.rf.external+recal	0.776	0.193	1.05	0.11	0.041
E2	depresjon-to-psykose.xgboost.external+recal	0.769	0.196	1.05	0.11	0.052
E2	psykose-to-depresjon.logreg.external+recal	0.689	0.254	0.35	0.99	0.185
E2	psykose-to-depresjon.rf.external+recal	0.728	0.221	0.77	0.67	0.129
E2	psykose-to-depresjon.xgboost.external+recal	0.671	0.258	0.49	0.93	0.200

159 Under honest internal validation the informative models were already miscalibrated (Table 5):
160 slopes of 0.53 (XGBoost) to 0.77 (random forest) in DEPRESJON and 0.95 (logistic regression)
161 in PSYKOSE, ECE 0.06 to 0.11 in DEPRESJON, and negative slopes with ECE 0.14 to 0.22 in
162 HYPERAKTIV, where the models had learned nothing but still produced confident probabilities.
163 Record-wise models were markedly *better* calibrated in-sample, which is itself a symptom: they
164 were scoring participants they had seen. Recalibrating external predictions with a slope and
165 intercept fitted on the training cohort’s own out-of-fold predictions repaired the easy direction
166 (DEPRESJON $\rightarrow$ PSYKOSE: slopes 1.05, ECE 0.04 to 0.05 for the ensembles) but not the
167 hard one (PSYKOSE $\rightarrow$ DEPRESJON: slopes 0.35 to 0.77, ECE 0.13 to 0.20), because the
168 miscalibration there is a property of the population shift, not of the model.

169 3.5 E4: sex-stratified performance

170 Discrimination did not differ materially by sex in DEPRESJON (AUROC 0.76 to 0.78 in both
171 strata) or HYPERAKTIV (chance in both). In PSYKOSE the female stratum contained three
172 cases, and its apparently higher AUROC (0.97) rests on those three participants; the intervals
173 overlap the male stratum’s (0.87 to 0.89). Calibration slopes differed more than discrimination
174 between strata (Table 6).

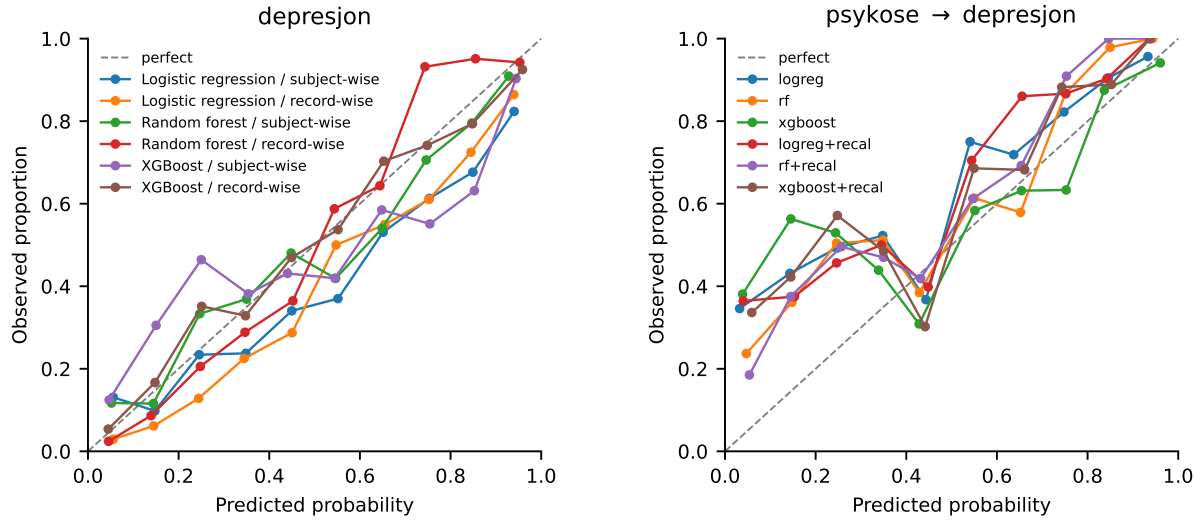


Figure 3: E3. Left: reliability curves of the subject-wise and record-wise E1 models on DEPRESJON. Right: external predictions PSYKOSE $\rightarrow$ DEPRESJON before and after logistic recalibration fitted on the training cohort.

Table 6: E4: sex-stratified performance of the subject-wise E1 models (window level, subject-level bootstrap intervals). Subgroups with fewer than six participants or a single class are omitted.

Cohort	Model	Sex	Participants (cases)	AUROC [95\% CI]	Brier	Cal. slope	ECE
depresjon	Logistic regression	female	30 (10)	0.777 [0.649, 0.872]	0.182	0.75	0.120
depresjon	Logistic regression	male	25 (13)	0.776 [0.562, 0.882]	0.192	0.62	0.091
depresjon	Random forest	female	30 (10)	0.763 [0.623, 0.870]	0.171	0.75	0.071
depresjon	Random forest	male	25 (13)	0.777 [0.626, 0.870]	0.193	0.80	0.070
depresjon	XGBoost	female	30 (10)	0.755 [0.620, 0.859]	0.183	0.51	0.109
depresjon	XGBoost	male	25 (13)	0.772 [0.619, 0.868]	0.203	0.55	0.120
hyperaktiv	Logistic regression	female	41 (21)	0.479 [0.370, 0.602]	0.275	-0.10	0.152
hyperaktiv	Logistic regression	male	44 (24)	0.424 [0.321, 0.543]	0.283	-0.49	0.168
hyperaktiv	Random forest	female	41 (21)	0.491 [0.371, 0.611]	0.269	-0.10	0.111
hyperaktiv	Random forest	male	44 (24)	0.398 [0.278, 0.516]	0.291	-0.80	0.160
hyperaktiv	XGBoost	female	41 (21)	0.498 [0.385, 0.605]	0.293	-0.02	0.211
hyperaktiv	XGBoost	male	44 (24)	0.388 [0.278, 0.505]	0.332	-0.44	0.230
psykose	Logistic regression	female	23 (3)	0.971 [0.915, 0.998]	0.096	1.58	0.177
psykose	Logistic regression	male	31 (19)	0.890 [0.771, 0.933]	0.126	0.83	0.081
psykose	Random forest	female	23 (3)	0.966 [0.891, 0.995]	0.075	1.66	0.147
psykose	Random forest	male	31 (19)	0.874 [0.765, 0.930]	0.149	0.90	0.123
psykose	XGBoost	female	23 (3)	0.969 [0.904, 0.997]	0.059	1.16	0.100
psykose	XGBoost	male	31 (19)	0.884 [0.774, 0.933]	0.140	0.71	0.121

3.6 E5: OBF-Psychiatric transdiagnostic and five-class arms

Table 7: E5: OBF-Psychiatric (162 participants, single control group). Transdiagnostic binary arm (any psychiatric group vs healthy control; window-level AUROC with subject-level BCa intervals) and five-class arm (macro one-vs-rest AUROC), each under subject-wise and record-wise CV.

Arm	Model	Split	AUROC / macro-AUROC	Accuracy	Macro-F1	Cal. slope
Any psychiatric vs control	Majority class	subject-wise	0.490 [0.358, 0.614]	0.641	0.391	0.25
Any psychiatric vs control	Majority class	record-wise	0.500 [0.470, 0.529]	0.641	0.391	0.25
Any psychiatric vs control	Logistic regression	subject-wise	0.795 [0.745, 0.848]	0.741	0.725	0.78
Any psychiatric vs control	Logistic regression	record-wise	0.824 [0.777, 0.867]	0.756	0.742	0.93
Any psychiatric vs control	Random forest	subject-wise	0.781 [0.726, 0.837]	0.739	0.705	0.75
Any psychiatric vs control	Random forest	record-wise	0.849 [0.811, 0.887]	0.782	0.755	1.07
Any psychiatric vs control	XGBoost	subject-wise	0.776 [0.729, 0.834]	0.734	0.702	0.63
Any psychiatric vs control	XGBoost	record-wise	0.844 [0.810, 0.885]	0.781	0.755	0.90
Five diagnostic groups	Logistic regression	subject-wise	0.713 [0.677, 0.750]	0.431	0.390	-
Five diagnostic groups	Logistic regression	record-wise	0.770 [0.740, 0.799]	0.484	0.444	-
Five diagnostic groups	Random forest	subject-wise	0.695 [0.662, 0.735]	0.434	0.350	-
Five diagnostic groups	Random forest	record-wise	0.811 [0.786, 0.843]	0.558	0.482	-
Five diagnostic groups	XGBoost	subject-wise	0.702 [0.670, 0.743]	0.448	0.355	-
Five diagnostic groups	XGBoost	record-wise	0.808 [0.785, 0.842]	0.558	0.467	-

In the harmonised 162-participant release, separating any psychiatric group from the healthy controls reached window-level AUROC 0.776 to 0.795 subject-wise and 0.824 to 0.849 record-wise (subject level 0.856 to 0.867 versus 0.899 to 0.943). Distinguishing the five diagnostic groups gave a macro one-vs-rest AUROC of 0.695 to 0.713 (accuracy 0.43 to 0.45 against a 0.28 majority rate) under subject-wise splitting and 0.770 to 0.811 (accuracy 0.48 to 0.56) under record-wise splitting (Table 7); the inflation was again largest for the tree ensembles (0.106 to 0.116 in macro-AUROC) and smallest for logistic regression (0.057).

3.7 Supplementary arm: 1D-CNN on the raw series

[TODO: One sentence from results/real/E1_cnn/e1_results.csv once the run completes.]

4 Discussion

4.1 Principal findings

Each practice the review found to be common changed conclusions on real data, always towards optimism. Record-wise splitting added 0.02 to 0.17 AUROC and turned a null result into an apparently useful classifier. Skipping external validation hid a loss of up to 0.20 AUROC and a collapse of calibration in one transfer direction while the other direction happened to hold, which is exactly why one cannot know without doing it. Calibration, never reported for these cohorts, was poor even under honest validation and was not repairable from the training data alone.

4.2 Relation to the published benchmarks

The honest subject-level AUROC of 0.86 to 0.89 on DEPRESJON is consistent with the dataset authors' leave-one-patient-out baseline [4] and well below the 0.95 to 0.98 that our own record-wise arm produces. Published accuracies above 0.9 on these cohorts [8, 9, 10] are therefore reproducible, but only under the design that the companion review scores as high risk of bias. The HYPERAKTIV result deserves emphasis: one day of activity counts carries no information

about ADHD versus other psychiatric outpatients in this sample, and any claim to the contrary that rests on day-level splits should be read as participant re-identification.

4.3 Limitations

Three cohorts from one device family and one country; small participant numbers with wide subject-level intervals; diagnostic-group labels rather than symptom trajectories; a single window length; models fixed a priori rather than tuned (tuning inside subject-wise folds would raise honest performance somewhat but would not change the paired inflation, which is the estimand); no demographic predictors; and, in E2, a control group split rather than an independent one, because no independent healthy control actigraphy with the same device is public. The sex analysis in PSYKOSE is underpowered.

4.4 Implications

Authors of ML models on repeated-measures mental-health data should state the unit of splitting, report subject-level as well as window-level sample sizes and metrics, report calibration slope and intercept with a reliability curve, and check for shared participants across the datasets they combine. Reviewers can ask for these in one sentence each. Journals could require them. The companion review reports how far the field is from that standard; this study reports what the distance costs.

Declarations

Funding: [TODO:]. Conflicts of interest: none declared. Ethics: secondary analysis of public, de-identified data; [TODO: institutional determination]. Registration: [TODO: OSF]. Data availability: DEPRESJON, PSYKOSE, HYPERAKTIV and OBF-Psychiatric are available from their authors [4, 5, 6, 7]; checksums in the repository. Code availability: <https://github.com/rogerpanel/cv/tree/main/mlmh-appraisal> ([TODO: tag; Zenodo DOI]).

Supplementary material

S1 TRIPOD+AI checklist (`supplement_tripod_checklist.tex`); S2 feature definitions (`src/mlmh/features/actigraphy.py`); S3 per-window predictions and run manifests (`results/real/`).

References

- [1] Ajibola AA, et al. Methodological quality and reporting completeness of machine learning models for mental health prediction: a systematic review and PROBAST+AI / TRIPOD+AI appraisal. [TODO: under review / preprint DOI]. 2026.
- [2] Moons KGM, Damen JAA, Nair T, et al. PROBAST+AI: an updated quality, risk of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods. *BMJ*. 2025;388:e082505.
- [3] Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement. *BMJ*. 2024;385:e078378.
- [4] Garcia-Ceja E, Riegler M, Jakobsen P, et al. Depresjon: a motor activity database of depression episodes in unipolar and bipolar patients. *ACM MMSys*. 2018. doi:10.1145/3204949.3208125
- [5] Jakobsen P, Garcia-Ceja E, Stabell LA, et al. PSYKOSE: a motor activity database of patients with schizophrenia. *IEEE CBMS*. 2020. doi:10.1109/CBMS49503.2020.00064

- 236 [6] Hicks SA, Stautland A, Fasmer OB, et al. HYPERAKTIV: an activity dataset from patients with attention-
237 deficit/hyperactivity disorder (ADHD). ACM MMSys. 2021. doi:10.1145/3458305.3478454
- 238 [7] Garcia-Ceja E, et al. OBF-Psychiatric, a motor activity dataset of patients diagnosed with major depression,
239 schizophrenia, and ADHD. Sci Data. 2025;12. doi:10.1038/s41597-025-04384-3
- 240 [8] Explainable AI for depression detection and severity classification from activity data. JMIR Ment Health.
241 2025;12:e72038.
- 242 [9] Transfer learning for real-time deployment of a screening tool for depression detection using actigraphy.
243 arXiv:2303.07847. 2023.
- 244 [10] Unveiling psychotic disorder patterns: a deep learning model analysing motor activity time-series data with
245 explainable AI. Biomed Signal Process Control. 2024; and Objective diagnosis of psychotic disorders using
246 multi-branch deep learning on motor activity signals. Soft Comput. 2024.
- 247 [11] Van Calster B, McLernon DJ, van Smeden M, et al. Calibration: the Achilles heel of predictive analytics.
248 BMC Med. 2019;17:230.
- 249 [12] Saeb S, Lonini L, Jayaraman A, Mohr DC, Kording KP. The need to approximate the use-case in clinical
250 machine learning. GigaScience. 2017;6:gix019.